

Kevin Rozmiarek

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Education

Ph.D., Earth Science, Department of Earth Science, University of Colorado, Boulder August 2026

Advisors: Dr. Tyler Jones and Dr. Irina Overeem

Thesis: *Revealing carbon cycle secrets in the Arctic: Biogeochemical indicators of methane escape from abrupt thaw permafrost lakes*

M.Sc., Applied Physics, Department of Physics, Colorado School of Mines May 2016

Advisor: Dr. Timothy Ohno

Thesis: *Numerical Optimization of Ion Transport in Electrospray Mass Spectrometer Sources*

B.Sc., Engineering Physics, Department of Physics, Colorado School of Mines May 2015

Advisor: Dr. Timothy Ohno

Senior Capstone: *Laser Ablation-Combustion Instrument Interface for Carbon Isotope-Ratio Mass Spectrometry*

Professional Research Appointments

Research Scientist I, Institute of Arctic and Alpine Research, CU Boulder August 2026 – Present

Advisor: Dr. Bradley Markle

Developing machine learning models trained on Lagrangian back trajectories coupled to water isotope distillation to inform hydroclimate in the Arctic.

Graduate Research Assistant, Institute of Arctic and Alpine Research, CU Boulder August 2020 – August 2026

Advisors: Dr. Tyler Jones and Dr. Irina Overeem

Understanding the carbon isotope flux of methane derived from newly formed thermokarst permafrost features and their contribution to the Arctic methane budget using isotope-enabled biogeochemical ecosystem models.

Professional Scientist, Institute of Arctic and Alpine Research, CU Boulder April 2017 – December 2020

Supervisor: Bruce Vaughn

Operated and maintained NOAA's Global Monitoring Division's measurement network for stable isotopes of methane and carbon dioxide.

Research Contractor at United States Geological Survey May 2015 – September 2016

Advisor: Dr. Andrew Hunt

Characterized the isotope signature of water samples across the noble gas spectrum to approximate aquifer recharge rate with the USGS Noble Gas Stable Isotope Lab.

Research Assistant at Colorado School of Mines/United States Geological Survey August 2014 – May 2015

Advisor: Dr. Matthew Emmons

Designed and built a laser-ablation and combustion-furnace sample preparation device for integration into a magnetic-sector isotope-ratio mass spectrometer in collaboration with United States Geological Survey.

Research Assistant at Colorado School of Mines August 2013 – May 2014

Advisor: Dr. Stephen Pankavich

Performed work as an Undergraduate Research Fellow in the Department of Applied Mathematics & Statistics. Developed a multi-compartmental population model for stage 1 HIV-1 infections within the gut associated lymphatic tissue, bronchus associated lymphatic tissue, and the peripheral blood

Teaching Appointments

Instructor at The University of Colorado Department of Aerospace Engineering

Taught **Graduate Projects**, a master's graduate course which delivers a master's capstone project for a group of students. Advised two different groups, one group of four and another group of ten. Wrote and delivered lectures on engineering project lifecycle and relevant Earth science topics. Guided student delivery on final engineering product for developing their professional portfolios.

August 2024 – December 2024

August 2025 – May 2026

Teaching Assistant at The University of Colorado Department of Geological Sciences

Supervisor: Dr. Laura Chimiak

Taught **Water, Energy and Environment: An Introduction to Earth Resources**, a freshman level course for non-majors. Focused on an even mixture of basic geology and modern policy topics relating to subject matter. Oversaw recitations for coursework. Included mini lectures.

August 2025 – May 2026

Assistant Teacher at The University of Colorado Department of Atmospheric and Oceanic Sciences

Taught an **Independent Study** for two different students. Developed students' ability for independent inquiry centered around the engineering of sensors for meteorological study. Both students joined field work with project deliverables.

August 2023 – December 2025

Teaching Assistant at The University of Colorado Department of Atmospheric and Oceanic Sciences

Supervisor: Dr. Andrew Winters

Taught **Weather and Atmosphere**, a freshman level course. Led course for non-majors introducing topics on the fundamentals of the atmosphere. Oversaw recitations for coursework, included mini lectures.

January 2025 – May 2025

Teaching Assistant at The University of Colorado Department of Atmospheric and Oceanic Sciences

Supervisor: Dr. Derek Brown

Taught **Weather and Atmosphere Lab**, a freshman level course. Led lab course for non-majors introducing topics on the fundamentals of the atmosphere. Created content, oversaw lab operation, and evaluated student performance. Included mini lectures.

January 2024 – May 2024

Teaching Assistant at Colorado School of Mines Department of Physics

Supervisor: Dr. Lawrence Wiencke

Taught **Advanced Physics Lab II**, a junior level course. Led lab course on nuclear and solid-state physics with an emphasis on writing. Oversaw lab operation and evaluated student performance. Included mini lectures.

January 2016 – May 2016

Teaching Assistant at Colorado School of Mines Department of Physics

Supervisor: Dr. Frank Kowalski

Taught **Advanced Physics Lab I**, a junior level course. Led lab course on optics and atomic physics with an emphasis on experimental design. Oversaw lab operation and evaluated student performance. Included mini lectures.

August 2015 – December 2015

Teaching Assistant at Colorado School of Mines Department of Physics

Supervisor: Dr. Timothy Ohno

Taught **Field Session Techniques in Physics**, a junior level course. Worked with students on labs on the fundamentals of vacuum science, gas transport, mass spectrometry, surface characterization, and thin film deposition. Designed curriculum, lectured students, oversaw lab operation, and evaluated student performance.

May 2014 – July 2014

Awards and Honors

- 2026 IPCCS Travel Grant (\$1500)
- 2026 George R. Aiken Graduate Fellowship (\$3000)
- 2025 Graduate School Fall 2025 Domestic Travel Grant (\$650)
- 2025 INSTAAR Graduate Student Community Award
- 2025 Geology Spring 2025 Travel Award (\$1000)
- 2025 U.S. Open Ice Core Meeting Travel Award (\$250)
- 2024 U.S. Open Ice Core Meeting Travel Award (\$300)
- 2023 Graduate School Fall 2023 Domestic Travel Grant (\$450)
- 2023 Best Poster – Front Range Isotope Day
- 2023 Visiting Scholar Stipend – University of Bergen (\$1350)
- 2023 U.S. Open Ice Core Meeting Travel Award (\$300)
- 2015 Best Senior Design – Colorado School of Mines Department of Physics

Publications

- In prep **Rozmiarek KS**, Town M, Morris V, Chase BM, Nunn R, Wainright S, Vaughn BH, Markle B, Jones TR, “Multidecadal effects of storage sublimation on ice cores”.
- Submitted Malone SL, Chen S, Liu L, Xi X, Yuan F, Zhuang Q, Oh Y, McNicol G, Zhu Q, Malhotra A, **Rozmiarek KS**, Arndt KA, Li F, Monteverde DR, Talib A, Ward EJ, Ying Q, Yuan K, Zheng J, “Drought raises wetland methane emissions only when it is hot” *AGU Advances*
- Submitted Chen S, Zhu Q, Liu L, Zheng J, Jia X, Metzger S, Oh Y, Arndt KA, Malhotra A, Malone SL, McNicol G, **Rozmiarek KS**, Smith CCR, Yuan F, Yuan K, Ma Y, Chen Z, Li F, Ward E, Yang Y, Ying Q, Zhuang Q, “Next-Generation Monitoring and Modeling of Wetland Methane Dynamics through Artificial Intelligence” *Nature Climate Change*
- In review Miller EA, Coburn S, Alden C, **Rozmiarek KS**, Henze D, Jones TR, Rieker GB, “Quantifying area-source methane fluxes with path-averaged laser beam concentration measurements and Bayesian inversion”. *Atmospheric Measurement Techniques*.
- In review **Rozmiarek KS**, Liu X, Oh Y, Overeem I, Miller EA, Morris V, Vaughn BH, Hasson NR, Chase BM, Walter-Anthony KM, Zhuang Q, Rieker GB, Jones TR, “Multi-model Insights into $\delta^{13}\text{C}$ -CH₄ from Arctic Permafrost Thermokarsts”. *JGR Atmospheres*.
- In review **Rozmiarek KS**, Miller EA, Overeem I, Rieker GB, Jones TR, “Evaluation of area-source methane-emission quantification by Uncrewed Aerial Systems using Gauss's law”. *Atmospheric Measurement Techniques*.
- In review Morris VA, Vaughn BH, Jones TR, Rasmussen SO, Vinther BM, Gkinis V, Skorski WB, Hughes AG, Brashear CA, Koldtoft I, Kirk M, Svensson AM, Behrens M, Conrad E, Diekstal D, Horhold M, Andreasen N, Born A, Holme CT, Jensen MF, Smith-Johnsen S, Kjellman SE, Langebroek PM, Mekhaldi F, Nunn RM, Rheinlaender JW, Siegel PH, Simon MH, Sinnl G, Woong J, Vimont AJ, Englund Michel S, Buizert C, Sowers T, Steen-Larsen HC, **Rozmiarek KS**, Nisancioglu KH, Steffensen JP, Dahl-Jensen D, White JWC, “An Ultra-High-Resolution Record of Water Stable Isotopes for the East Greenland Ice-Core Project” *Scientific data*
- 2025 **Rozmiarek KS**, Schamback J, Bennett H, Yang J, Ricken B, Eberling N, Mays W, Caro T, Jones T, Smallwood C, “Microbial and Chemical Predictors of Methane Release from a Stratified Thermokarst Permafrost Hotspot” *Frontiers in Microbiology*.

- 2025 **Rozmiarek KS**, “Collecting atmospheric air with drones” *Nature Reviews Earth & Environment* [Non-peer reviewed article]
- 2025 **Rozmiarek KS**, Dietrich LJ, Vaughn BH, Town MS, Markle B, Morris V, Steen-Larsen HC, Fettweis X, Brashear CA, Bennett H, Jones TR, “Atmosphere to surface profiles of water vapor isotopes and meteorological conditions over the northeast Greenland ice sheet” *JGR Atmospheres*.
- 2025 Brashear CA, Jones TR, Morris V, Vaughn BH, Robert WHG, Skorski WB, Hughes AG, Nunn R, Ramussen SO, Cuffey KM, Vinther BM, Sowers T, Buizert C, Gkinis V, Holme C, Jensen MF, Kjellman S, Langebroek PM, Medkhaldi F, **Rozmiarek KS**, Rheinlænder JW, Simon M, Sinnl G, Smith-Johnsen S, White JWC, “Shifts in Greenland interannual climate variability lead Dansgaard-Oeschger abrupt warming by hundreds of years”. *Climate of the Past*.
- 2023 Jones TR, Cuffey KM, Roberts WHG, Markle BR, Steig EJ, Steven, CM, Valdes PJ, Fudge TJ, Sigl M, Hughes AG, Morris V, Vaughn BH, Garland J, Vinther BM, **Rozmiarek KS**, Brashear CA, White JWC, (2023) “Seasonal temperatures in West Antarctica during the Holocene”. *Nature*.
- 2021 **Rozmiarek KS**, Vaughn BH, Jones TR, Morris V, Skorski WB, Hughes AG, Elston J, Wahl S, Faber A-K, Steen-Larsen HC, (2021) “An unmanned aerial vehicle sampling platform for atmospheric water vapor isotopes in polar environments”. *Atmospheric Measurement Techniques*.

Selected Talks

- Sep 2023 **Sandia Day 2023**, Boulder – *Progress towards boreal isotope-enabled methane biogeochemical modeling*
- Aug 2023 **International Society for Atmospheric Research using Remotely piloted Aircraft Annual Meeting**, Bergen – *Whole-air sampling by uncrewed aircraft enables investigation of isotopic composition of water vapor in polar environments*
- April 2023 **CU's 17th Annual Hydrosociences Symposium**, Boulder – *Atmosphere to surface profiles of water vapor isotopes and meteorological conditions over the northeast Greenland ice sheet reveals the role of sublimation to the ice sheet hydrological cycle*
- Oct 2022 **University of Colorado Department of Geological Sciences Colloquium**, Boulder – *Uncrewed aircraft sampling of atmospheric water-vapor isotopes at a Greenland ice core camp*
- Aug 2022 **Willi Dansgaard Centenary Symposium**, Copenhagen – *Uncrewed aircraft sampling of atmospheric water-vapor isotopes at a Greenland ice core camp*
- Aug 2019 **Front Range Isotope Day (FRIday)**, Boulder – *Flying Higher: An update to INSTAAR's water vapor isotope drone program and what it could mean for understanding the Greenland hydrological cycle*

Field work

- 2025 Navigating the New Arctic (summer) – **Field leader (oversaw team of 4)** – Fairbanks/Akiachak/Fort Yukon, AK
- 2024 Navigating the New Arctic (summer) – **Field leader (oversaw team of 8)** – Fairbanks, AK
- 2023 Navigating the New Arctic (summer) – **Field leader (oversaw team of 6)** – Fairbanks, AK
- 2023 Navigating the New Arctic (spring) – **UAS Pilot** – Fairbanks, AK
- 2022 Navigating the New Arctic (fall) – **UAS Pilot** – Fairbanks, AK
- 2022 EGRIP Ice Core Camp – **UAS team lead (oversaw science team of 4)** - Northeast Greenlandic Ice Stream
- 2019 EGRIP Ice Core Camp – **UAS team lead (oversaw science team of 2)** - Northeast Greenlandic Ice Stream
- 2017 WAIS Divide Ice Core Reprocessing - **SIL Processing Line Manager (oversaw 6 undergraduates)** – National Science Foundation Ice Core Facility, Lakewood, Colorado
- 2015 Meeker aquifer sampling – **Technician** – Meeker, Colorado, USA

Service and Outreach

KEY: CU (University of Colorado, Boulder), INSTAAR (Institute of Arctic and Alpine Research), CSM (Colorado School of Mines)

Review – Journal Reviewer – <i>The Cryosphere</i> , Copernicus	2025
Review – Journal Reviewer – <i>Atmospheric Chemistry and Physics</i> , Copernicus	2025
INSTAAR – INSTAAR Student Gear Closet – Started and operates a funding pool and resource for students to receive Arctic field gear based on student needs	2024 – Present
CU – UAS Advisory Committee (UAC) – Serves in an advisory role to the Vice Chancellor for Public Safety Operations and Director of Flight Operations	2024 – Present
CU – Graduate Student Representative to the Geology Faculty – Representative of graduate students to academic department	2024 – 2025
INSTAAR – Search Committee for the Director of INSTAAR – Appointed by CU Research and Innovation Office to represent graduate students	2024
INSTAAR – INSTAAR Directorate Non-voting Member – Representative of graduate students on the governing body of the institute	2023 – 2025
INSTAAR – Methane Study Group – Led weekly journal club on selected methane topics	2021 – 2023
INSTAAR – INSTAAR Open House – Annually gives tours to middle school students on the role INSTAAR has in climate science	2017 – Present
CSM – Graduate Student Government – Physics department acting chair	2016
CSM – Equality through Awareness – Helped implement mentorship programs for underrepresented identity groups within the Colorado School of Mines Physics department.	2014 – 2016
CSM – Society of Physics Students - Little Shop of Physics – Visited a dozen low-income secondary schools introducing students to physics topics as a presenter and demonstrator.	2012 – 2015

Media Coverage of Research Activities

2026	(Main feature) <i>Ages of Ice</i> – PBS – https://www.pbs.org/show/ages-of-ice/
2025	(Interview) <i>The College Where Drones Are Everywhere</i> – The Chronicle of Higher Education – https://www.chronicle.com/article/the-college-where-drones-are-everywhere
2025	(Interview) <i>Drone experiment reveals how Greenland ice sheet is changing</i> – CU Boulder Today – https://www.colorado.edu/today/2025/03/27/drone-experiment-reveals-how-greenland-ice-sheet-changing
2023	(Appearance) <i>The Arctic's Permafrost-Obsessed Methane Detectives</i> – Magazine Article – WIRED – https://www.wired.com/story/arctic-permafrost-obsessed-methane-detectives/
2023	(Interview) <i>På tykk is</i> – Documentary – NRK TV – https://tv.nrk.no/serie/paa-tykk-is/
2019	(Appearance) <i>Ice on Fire</i> – Documentary – HBO – https://en.wikipedia.org/wiki/Ice_on_Fire_(2019_film)

Et Cetera

Professional Memberships: International Permafrost Association, Permafrost Young Researcher Network, Alaska Geological Society, International Society for Atmospheric Research using remotely piloted Aircraft, ESIL AI for Natural Methane Working Group (Perspectives sub-working group leader), American Geophysical Union, European Geosciences Union

Modeling Skills: Running large scale Earth System Models on high performance computing clusters (Terrestrial Ecosystem Model, isoTEM); I hold a software licensing agreement for payload control software with Black Swift Technologies; I have worked professionally using Python, MATLAB, R, Mathematica, React, Typescript, C++, Lua, and Java.

Software Tools: Autodesk Fusion, EAGLE, SolidWorks, LabVIEW, Inkscape, SIMION, PIX4Mapper, Microsoft Office Suite

AI Tools: Claude Code, Codex, Midjourney, Kling, Perplexity, ChatGPT

Laboratory Skills: Isotope ratio mass spectrometry including magnetic sector, quadrupole, and ion trap; gas chromatography with combustion, pyrolysis, FID, TCD; vacuum technology; cavity ring-down spectroscopy

Fabrication and design: Machine shop fabrication; embedded system design, circuit and PCB design; 3D printing